

Aldehydes and ketones

By: Assist. Lecturer

Sazgar A. Hameed

MSc. in Clinical Biochemistry

College of Medicine – HMU

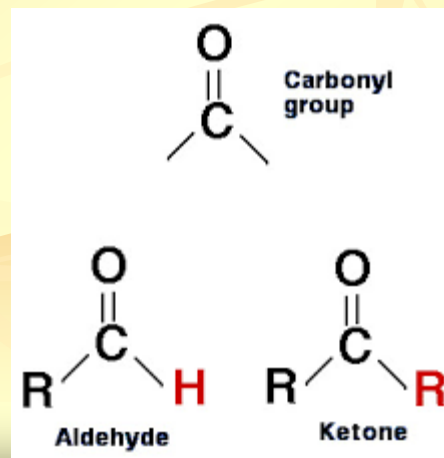
Aldehydes and ketones

Aldehydes and Ketones are two classes of compounds that contain the carbonyl group.

Examples:

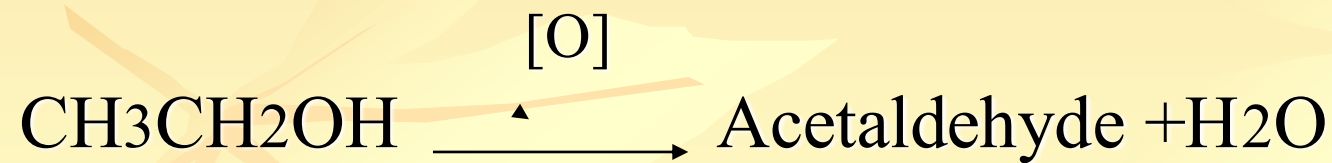
Aldehydes: Acetaldehyde, Benzaldehyde.

Ketones: Acetone, Acetophenone, Benzophenone



Preparing Aldehydes and Ketones

Oxidation of primary alcohols produces aldehydes.



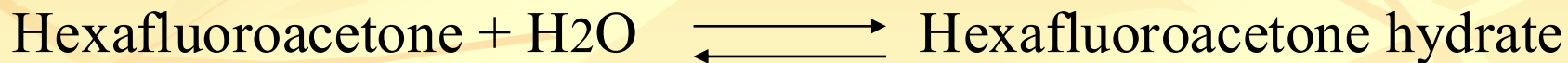
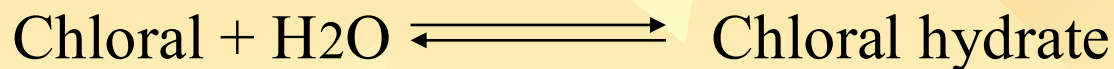
The oxidation of secondary Alcohols yields a ketone.



Reactions of Aldehydes and Ketones

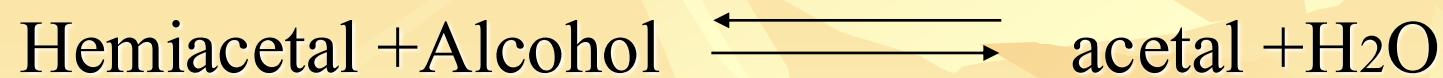
1-Addition reactions

A-Addition of water

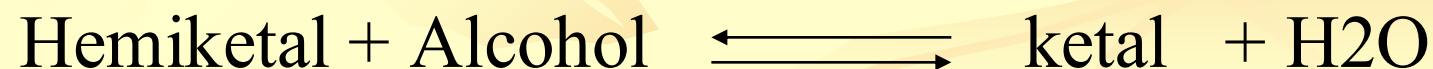
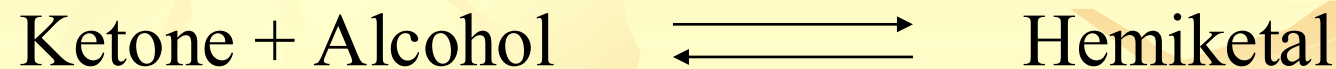


Chloral hydrate is an example of a stable hydrate. It is a crystalline compound that is used in veterinary medicine as an anesthetic for animals.

B-Addition of alcohols (formation of hemiacetal, acetal, hemiketal, and ketal)

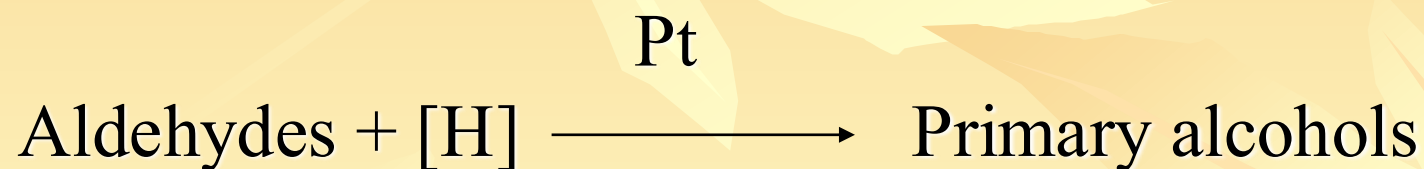


A hemiacetal contains an alkoxy(OR) and alcohol (OH) group, both bonded to the original carbon of the carbonyl group.

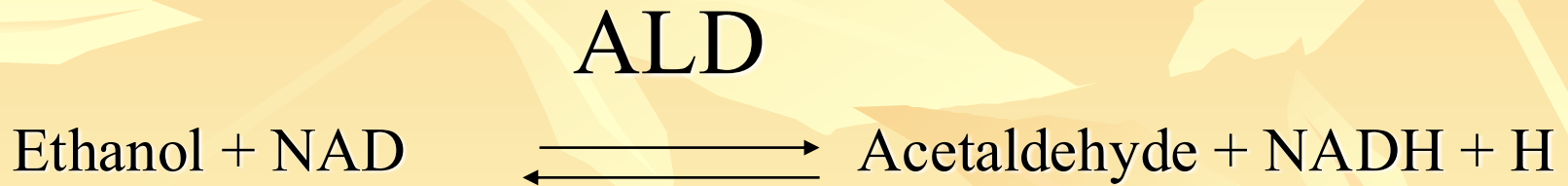


2- Reduction of aldehydes and ketones

The reduction of aldehydes and ketones forms alcohols. primary alcohols are formed from aldehydes, whereas secondary alcohols are formed from ketones.



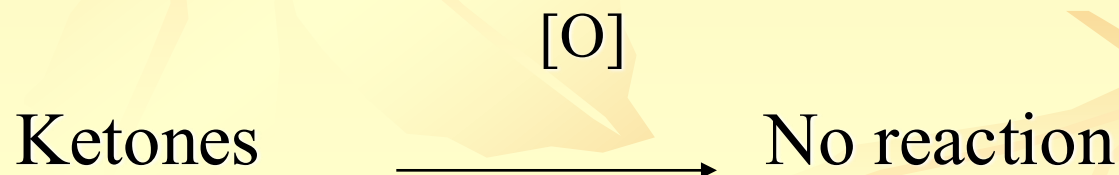
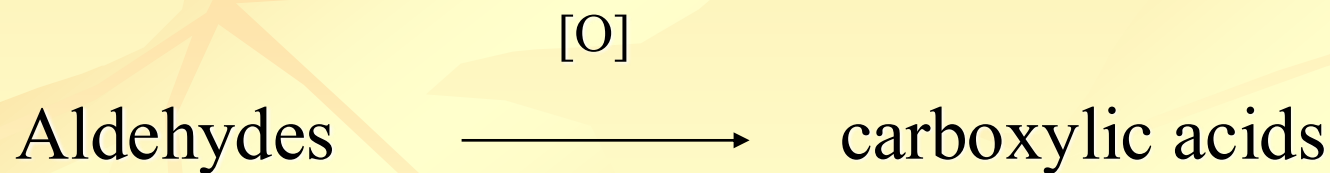
Biological oxidation of alcohols also occur in the body.



ALD = Alcohol dehydrogenase

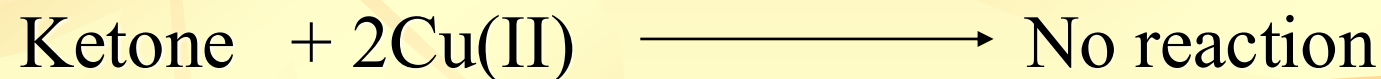
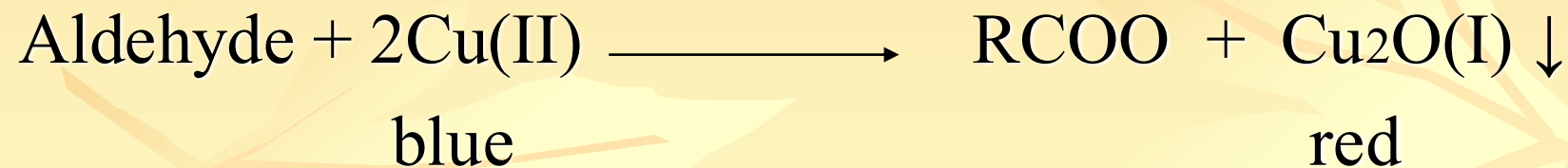
3-Oxidation reactions of Aldehydes

- Although aldehydes and ketones are similar in structure and generally show the same chemical properties, they differ in their ability to undergo oxidation.
- Aldehydes are easily oxidized, but ketones are not; this is because the C-H bond is **more reactive** than the C-R bond.

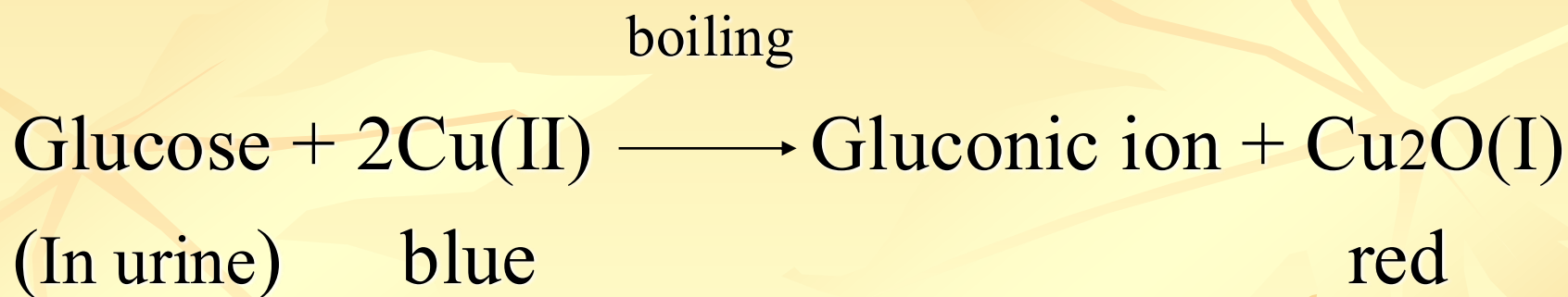


- Oxidation is the basis of several chemical tests used to distinguish an Aldehyde from a Ketone. Two tests commonly used in the laboratory are **Tollen's** test and **Benedict's** test.

- Benedict's reagent is a solution of copper (II) ion in a complex with citrate ions, when benedict's reagent is added to compound containing an aldehyde group, the aldehyde is oxidized to the carboxylic ion, and the copper (II) ion, which is blue color is reduced to the copper (I) oxide, which is red color and precipitates due to it is low solubility.



- Benedict's test is used in clinical laboratories to test for the presence of glucose in urine, so that Benedict's test is used to diagnose diabetes mellitus, because in a normal person, glucose does not appear in urine.



4-Aldol condensation

The condensation reaction involves the addition of one aldehyde to another, with bases as a catalyst.



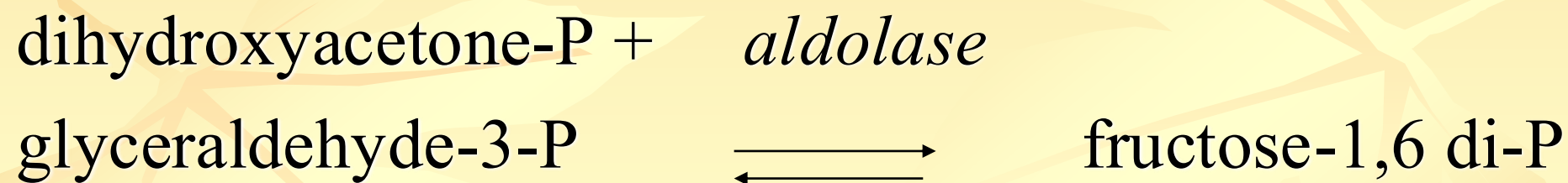
In the condensation reaction, two important things happen.

1-The carbon chain is increased.

2-The product has an aldehyde group and also an alcohol group; this type of condensation is called an aldehyde condensation.

Condensation reactions in living systems

Carbohydrates are synthesized in living systems by an enzyme-catalyzed aldol condensation.



Medical uses of aldehydes and ketones

1- Aldehydes

A. Formaldehyde (Methanal)

Formaldehyde is a colorless gas with a very sharp odor. It is used in the laboratory as a water solution containing 40%, commonly known as **formalin**, which is an effective germicide for the disinfection of rooms and clothing. It is also used as a preservative for biological specimens. Formaldehyde solutions shouldn't be used directly in the room with the patient because of irritating fumes.

B. Glutaraldehyde

- Glutaraldehyde is a sterilizing agent, and it's also microcidal against all microorganisms, including spores and viruses.
- Glutaraldehyde does not have a disagreeable odor like formaldehyde, and it's less irritating to the eyes and skin.

C. Glycer aldehyde

Is an important component in the metabolism of carbohydrates.

2-Ketones

Acetone (Propanone)

Acetone is an important product of lipid metabolism in the body; its concentration in the blood increases in diabetes mellitus. In diabetes, glucose utilization is inhibited, and there is excessive breakdown of fats, since acetone is one of the normal products of fat metabolism, is excreted in the urine. As the acetone concentration of the blood increases, the volatile acetone is exhaled; this is a characteristic sign of diabetes.

Aromatic aldehydes

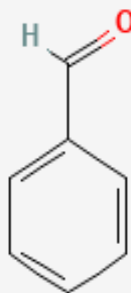
Some aromatic aldehydes are used to flavor food and as fragrances in perfumes.

Examples:

- Benzaldehyde is found in Almonds
- Vanillin in Vanilla beans
- Cinnamaldehyde in Cinnamon.

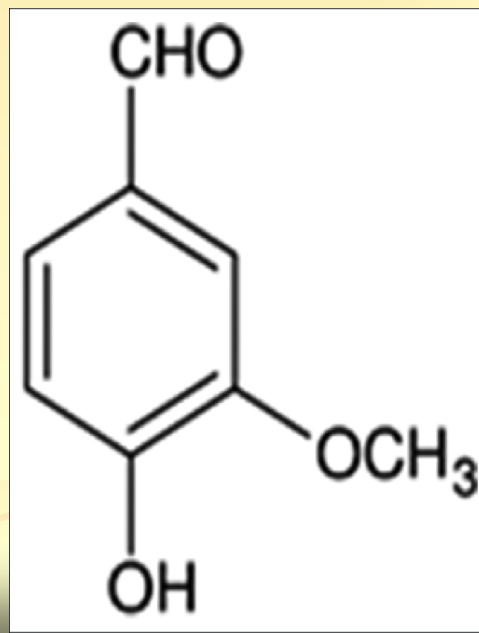
Benzaldehyde

Is an organic compound consisting of a benzene ring with a formyl substituent. It is the simplest aromatic aldehyde and one of the most industrially useful. It is a colorless liquid with a characteristic almond odor.



Vanillin

Is a phenolic aldehyde; its functional groups include aldehyde, hydroxyl, and ether. It is the primary component of the extract of the vanilla bean. Synthetic vanillin is now used more often than natural vanilla extract as a flavoring agent in foods, beverages, and pharmaceuticals.



Cinnamaldehyde

It is an organic compound and a phenylpropanoid. It gives cinnamon its flavor and odor. This pale yellow, viscous liquid, which occurs in the bark of cinnamon trees and other species of the genus *Cinnamomum*, is cinnamaldehyde. The essential oil of cinnamon bark is about 90% cinnamaldehyde.

